Governed Semantic Tokens: Tokenization as an Admissibility Boundary

Tokenization as the Jurisdiction Where Meaning Is Allowed to Count

GST replaces opaque, model-specific vocabularies with registry-backed token identifiers whose semantics are explicit, typed, versioned, and governed.

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Abstract

Tokenization is typically treated as a preprocessing convenience: a lossy compression step that converts inputs into model-consumable units. While effective for scaling, this framing obscures a decisive systems question: **where does meaning become admissible at all?** This paper introduces **Governed Semantic Tokens (GST)**, a framework that reframes tokenization as a **jurisdictional boundary**—the point at which observations, language, and actions are promoted from raw signals into units of lawful semantic effect.

GST replaces opaque, model-specific vocabularies with **registry-backed token identifiers** whose semantics are explicit, typed, versioned, and governed. Tokens are no longer merely units of prediction; they are **units of admissible meaning**. Each token carries constraints on provenance, authority, and permissible transitions, enabling refusal, auditability, and semantic stability across models, modalities, and time.

Central claim. When tokenization is treated as an admissibility boundary rather than a compression trick, systems exhibit higher coherence, structural resistance to injection, cross-model semantic stability, and auditable control over promotion to action—without requiring access to model internals or assumptions about cognition. Governance moves to the boundary where meaning enters the system, replacing trust with compilation.

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1. Scope and Motivation

As language models expand into multimodal, tool-using, and agentic settings, failures increasingly occur before or after inference rather than during it. Hal-

lucination, prompt injection, authority leakage, and silent escalation are rarely failures of fluency. They are failures of **promotion**: text, images, or events are allowed to count when they should not.

Tokenization is the earliest shared interface between messy reality and model cognition. Yet it remains ungoverned. Tokens inherit meaning implicitly from training corpora and drift across models, versions, and deployments. There is no principled place to say: *this may be observed, but not acted upon.*

This paper argues that tokenization is that place.

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2. From Tokens as Compression to Tokens as Law

Conventional tokenizers optimize for throughput, reuse, and reconstruction. Their success criteria are numerical; their semantics are implicit. Meaning emerges indirectly through training and is inseparable from a particular model family.

Governed Semantic Tokens invert this priority:

- tokens are explicit semantic objects rather than latent codes,
- meaning is declared and enforced outside the model,
- the tokenizer functions as a compiler from signals into governed identifiers.

Under GST, tokenization answers not "how should this be encoded?" but "**what is this allowed to mean?**"

This inversion does not change how models generate sequences. It changes what the surrounding system is permitted to treat as real.

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3. The Governed Semantic Token

A Governed Semantic Token is a numeric identifier backed by a registry entry. The registry defines the token's lawful semantics and admissible use. A minimal illustrative entry may resemble:

```
token_id: 4723
{
  type: "claim",
  provenance: { source: "ocr", confidence: 0.92 },
  authority: "observer",
  admissible_transitions: ["observe", "propose", "refuse"],
  version: "1.1"
}
```

This structure is illustrative, not prescriptive. Registries may be richer or sparser, centralized or federated. What matters is that semantics are explicit, inspectable, and stable across time.

Multiple surface forms—languages, templates, or visual realizations—may compile to the same token when they assert the same governed object. Conversely, identical surface text may compile to different tokens depending on provenance, type, or authority.

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4. Tokenization as an Admissibility Boundary

Let raw inputs (text, pixels, events) belong to a signal space X. Let governed tokens belong to a registry R.

The tokenizer is a partial compiler:

$$\tau : X \to R \cup \{\bot\}$$

If $\tau(x) = r$, the signal is admitted as a governed token with declared semantics. If $\tau(x) = \bot$, the signal is visible but inadmissible: it may be displayed or summarized, but it cannot participate in binding conclusions or actions.

This undefinedness is not an error. It is the semantic basis of refusal at the boundary. The system may reason *about* inadmissible signals, but it may not reason *with* them as authority-bearing units.

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5. Multimodality Without Latent Ambiguity

Recent multimodal systems demonstrate that images, audio, and video can be compressed into discrete token streams for efficiency. These tokens, however, remain opaque latents whose semantics are learned and model-bound rather than declared.

GST treats multimodality differently. Modality-specific encoders compile signals into typed semantic identifiers:

- visual regions into object or field tokens,
- OCR outputs into untrusted observation tokens,
- UI events and sensor readings into event tokens.

All modalities converge into a single governed identifier space. The model reasons over a unified semantic stream whose meaning is fixed by registry law rather than latent alignment.[1]

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6. Injection Resistance by Construction

Prompt injection exploits the fact that traditional tokenizers do not distinguish data from instruction. Strings originating from untrusted sources are tokenized identically to system directives.

GST prevents this structurally:

- tokens are typed at the boundary,
- provenance is mandatory for instruction-capable tokens,
- flow constraints prevent observation tokens from crossing into action or syscall domains without explicit elevation.

Injection is no longer an adversarial string problem. It becomes a type violation detected before inference or promotion.

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7. Cross-Model Semantic Stability

In contemporary systems, token meaning is entangled with model weights and vocabulary files. Upgrading models subtly alters semantics, even when interfaces appear stable.

GST decouples semantics from models:

- the registry defines meaning,
- models approximate over the same identifier space,
- swapping models does not redefine what a token denotes or permits.

This enables heterogeneous model fleets, long-lived deployments, and offline/online interoperability without semantic drift.

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8. Refusal as Boundary Success

When a signal cannot be compiled into a governed token, the tokenizer refuses. This refusal is a successful enforcement of semantic law.

Two refusal modes are distinguished:

1. **Signal refusal:** insufficient provenance, ambiguity, or confidence.
2. **Authority refusal:** correct signal, insufficient permission to promote.

Both preserve system integrity by preventing illegitimate meaning from entering the semantic stream.

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9. Provenance and Audit at Token Granularity

Each admitted token carries a provenance envelope describing its source, transformation chain, confidence, and authority context. Audit logs thus become graphs of semantic identifiers with attached receipts, not opaque text transcripts.

Post hoc analysis no longer asks "what did the model say?" but "**which admissible tokens supported this outcome, and under what authority?**"

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10. Compression and Measurable Effects

GST does not abandon efficiency; it relocates it to the semantic layer.

Semantic deduplication collapses many surface realizations into a single governed identifier, suggesting order-of-magnitude reductions in effective token diversity for reasoning and planning. Type-level separation structurally eliminates entire classes of injection pathways, reducing exploit surfaces by construction rather than detection.

These effects arise from architecture, not tuning. They are properties of where meaning is admitted, not how probabilities are assigned.

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11. Relationship to MCIR

Governed Semantic Tokens can be understood as **MCIR at the tokenizer layer**.

Where MCIR governs proposals, state transitions, and syscalls, GST governs **admission of meaning itself**. The registry corresponds to MCIR's semantic substrate; refusal semantics align; promotion to action mirrors syscall elevation. In this sense, GST functions as *MCIR-0*: the minimal viable substrate upon which higher-order governance can operate.

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12. GST Pipeline (Illustrative)

The GST pipeline operates in stages:

1. **Raw Signals** -> Modality-specific encoders
2. **Proposal Tokens** -> Admissibility compiler (tau)
3. **Admission Decision** -> Admit or Refuse
4. **Governed Tokens (R)** -> Model inference
5. **Token Sequences** -> Promotion rules
6. **Action Layer** -> Execute or Refuse

Example Flows:

Flow 1 (Accepted): Pixels -> [signature_box] -> GST:4723 -> reasoning -> [syscall_scan] -> execute

Flow 2 (Refused at boundary): Untrusted text -> [instruction | REFUSED] -> log/display only

Flow 3 (Refused at promotion): Observation token -> reasoning -> [action_request | REFUSED] -> refuse

The diagram emphasizes that refusal is not terminal failure but lawful termination at multiple stages.

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13. What GST Is Not

GST is not a claim about model cognition, a replacement for training, or an attempt to make semantics decidable. It does not eliminate uncertainty or disagreement.

It governs permission, not truth.

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14. Failure Modes

GST may fail through:

- overconstraint leading to excessive refusal,
- registry sprawl diluting shared semantics,
- governance capture encoding bias,
- boundary leakage from ungoverned subsystems.

These failures are visible by design. Governance does not disappear; it becomes inspectable.

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15. Conclusion

Tokenization has been treated as plumbing. This paper argues it is constitutional.

By relocating governance to the boundary where meaning enters the system, Governed Semantic Tokens provide a place to say no—cleanly, early, and audibly—before language hardens into action. They do not make models smarter. They make systems honest about what is allowed to count.

As language increasingly acts on the world, the admissibility boundary is the deepest layer worth governing.

[1] For example, visual tokens produced by systems such as DeepSeek-OCR can be treated as typed observation identifiers rather than raw latent codes, enabling explicit refusal and controlled promotion.

Corpus-Wide Reference Positioning

This paper argues that tokenization is typically treated as a preprocessing convenience: a lossy compression step that converts inputs into model-consumable units. Its nearest external comparison fields are token, semantic, formal. The sources in [E1], [E2], [E3], [E4], [E5], [E6], [E7] are cited as predecessor literature, adjacent standards, or counterpressure - not as evidence that they caused the Ableman framework or that they validate its novel claims. The comparison isolates where established work already supplies provenance, formal assurance, institutional constraint, empirical method, or engineering practice, and where this paper adds a distinct claim about admissibility, authority, semantic custody, or governed realization.

Within the Ableman corpus, the highest-priority current cross-links are [A1], [A2], [A3], [A4], [A5], [A6], [A7]. These are links among separately custodied works, not silent identity mergers. Their role is to expose mutual dependencies, terminology collisions, and places where a reader should traverse the corpus before treating this paper as self-sufficient.

Reference-pass counterevidence and limit search

The pass applied the following pressure tests: Statistical compression can improve efficiency while destroying distinctions needed for interpretation; semantic labels can conversely overstate stability that usage does not support. Formal vocabularies and graph links make commitments explicit but cannot by themselves establish correct identity, shared interpretation, or lossless translation between contexts. Formal assurance remains relative to the encoded model, trusted base, and refinement chain; proof does not remove specification error or unmodeled environmental behavior. These limits are not generic disclaimers: they identify the assumptions that a later empirical, formal, or implementation review must try to break.

Reference-pass influence and novelty boundary

No newly added external source is asserted to be a documented authorial influence unless the predecessor source already said so. `Predecessor` means earlier relevant work;

`adjacent` means a useful comparison; `standard` means an interoperability or assurance baseline; and `counterevidence` marks work that narrows or pressures a claim. Novelty therefore remains a reviewable, claim-level question rather than an inference from a low similarity score or an absent citation.

References

External scholarship, standards, and prior art

[**E1**] Rico Sennrich, Barry Haddow, and Alexandra Birch. "Neural Machine Translation of Rare Words with Subword Units." *ACL 2016*, pp. 1715-1725. DOI: 10.18653/v1/P16-1162. https://doi.org/10.18653/v1/P16-1162 **Relationship:** Predecessor subword tokenization method used as a contrast to semantically governed token boundaries.

[**E2**] Taku Kudo and John Richardson. "SentencePiece: A Simple and Language Independent Subword Tokenizer and Detokenizer for Neural Text Processing." *EMNLP 2018: System Demonstrations*. DOI: 10.18653/v1/D18-2012. https://doi.org/10.18653/v1/D18-2012 **Relationship:** Predecessor tokenizer implementation used to separate statistical segmentation from semantic admissibility.

[**E3**] Peter D. Grunwald. *The Minimum Description Length Principle*. MIT Press, 2007. mitpress.mit.edu source **Relationship:** Predecessor theory of compression and model selection; compression is not assumed to preserve governance or meaning.

[**E4**] Alistair Miles and Sean Bechhofer, eds. *SKOS Simple Knowledge Organization System Reference*. W3C Recommendation, 18 August 2009. https://www.w3.org/TR/skos-reference **Relationship:** Adjacent standard for concepts, labels, schemes, notes, and semantic mappings.

[**E5**] Richard Cyganiak, David Wood, and Markus Lanthaler, eds. *RDF 1.1 Concepts and Abstract Syntax*. W3C Recommendation, 25 February 2014. https://www.w3.org/TR/rdf11-concepts/ **Relationship:** Adjacent graph data model and identity baseline.

[**E6**] W3C OWL Working Group. *OWL 2 Web Ontology Language Document Overview*, Second Edition. W3C Recommendation, 11 December 2012. https://www.w3.org/TR/owl2-overview/ **Relationship:** Adjacent formal ontology standard used to compare declared semantics and inference commitments.

[**E7**] Mark D. Wilkinson et al. "The FAIR Guiding Principles for Scientific Data Management and Stewardship." *Scientific Data* 3, 2016, 160018. DOI: 10.1038/sdata.2016.18. https://doi.org/10.1038/sdata.2016.18 **Relationship:** Predecessor stewardship principles for findability, accessibility, interoperability, and reuse.

Ableman corpus

[**A1**] Adam Ableman. *Token Parsimony*. Ableman Corpus current edition; work identity `PAPER.PUB.20382825`;
source SHA-256
`EC3AA22D32E2BBA21CFCB6D1FA2AE6EA8AC11B7781D19C51530A2F6D924B2E91`.

Corpus relationship: priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

[**A2**] Adam Ableman. *Constitutional Legends.* Ableman Corpus current edition; work identity `PAPER.PUB.20330324`;
source SHA-256
`440CE8A83C2F86C020D1E9E482759F71D08F694F1C2250B5B624AEA0A4237AA7`.
Corpus relationship: priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

[**A3**] Adam Ableman. *Governed Controlled Natural Language (GCNL).* Ableman Corpus current edition; work identity
`AMCS-LOCAL-WORK-99C179FB694070D1`;
source SHA-256
`75A4AE13167785A0761E1F9CB67A9975138166A7B2B06B4FDF37C1A895178193`.
Corpus relationship: priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

[**A4**] Adam Ableman. *Global Term Key: Identity Primitives in Governed Knowledge Systems.* Ableman Corpus current edition; work identity
`AMCS-LOCAL-WORK-358AC09F5A13587E`;
source SHA-256
`AFB29CFCB4B58A78B79709AD453CEE9088CCB2AE40F30A70260F1B54F5160579`.
Corpus relationship: priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

[**A5**] Adam Ableman. *Meaning as Binding.* Ableman Corpus current edition; work identity
`AMCS-LOCAL-WORK-5F976917F0B283EA`;
source SHA-256
`A470D1A0B9ED7C3CEAB6BB88BE4B642A2BA36427447F787764329FD574DA0C97`.
Corpus relationship: priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

[**A6**] Adam Ableman. *Governed Meaning Base (GMB).* Ableman Corpus current edition; work identity
`AMCS-LOCAL-WORK-D95CCC4FB7B58BFB`;
source SHA-256
`A3B62E50F8BEE125FDAE0D38C2839AFD1CCC138296268BF93217481E38AA48EA`.
Corpus relationship: priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

[**A7**] Adam Ableman. *Semantic Manifold Topology.* Ableman Corpus current edition; work identity `PAPER.PUB.20257244`;
source SHA-256
`B8F28CC652A73A3EE4CF6FD0E23C1ABC922A6660319B2FB4CC1CCAAA606EC64A`.
Corpus relationship: priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

Corpus source and companion formats: https://github.com/AblemanMosaic/ableman-corpus
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